

PROJECT LOCATION IS AS FOLLOWS:

UNICOR, Federal Prison Industries, Inc.

Butner Optical – Medium II

2 Old 75 Hwy

Butner, NC 27509

Request for Information (suppliers sought) is for market research and pricing for various materials for plastic injection molding. See section B for Materials

Please submit pricing in writing to Chad Privett at chad.privett2@usdoj.gov

All communication must be in writing. No phone calls will be accepted.

For a list, description and total estimated quantities of all items, refer to section B.

UNICOR's desired delivery is fifteen (15) days or sooner after issuance of each delivery order.

All pricing must include shipping costs.

THE FOLLOWING INFORMATION IS REQUIRED TO BE COMPLETED BY THE OFFEROR:

SAM UNIQUE ENTITY ID NUMBER:

VENDOR'S POINT OF CONTACT:

VENDOR'S TELEPHONE & FAX NUMBER:

VENDOR'S EMAIL ADDRESS:

VENDOR'S BUSINESS SIZE (i.e. SMALL, LARGE, ETC.):

Vendors must have an active registration in SAM.gov when submitting pricing.

THE SPECIFICATIONS ARE SUGGESTED. SIMILAR SPECIFICATIONS MAY BE ACCEPTABLE

MUST MEET FDA FOOD GRADE MATERIALS (FDA 21 CFR 177.1520)

SECTION B**SUPPLIES AND PRICES/COSTS**

<u>Item No.</u>	<u>Supplies/Service</u>	<u>Quantity</u>	<u>U/M</u>	<u>Unit Price</u>	<u>Total Price</u>	<u>ARO Date</u>
0001	CHG2198	10,000	LBS			

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Food-contact acceptable (FDA 21 CFR 177.1520)

Fast cycle injection molding performance.

3. Physical Properties

Property	Value	Test Method
Density	0.900 g/cm ³	ASTM D1505
Melt Flow Rate (230°C/2.16 kg)	20 g/10 min	ASTM D1238

These values indicate a medium flow PP copolymer suitable for high speed injection molding.

4. Mechanical Properties

Property	Value	Test Method
Tensile Strength (Yield)	24.1 MPa	ASTM D638
Tensile Elongation (Yield)	9%	ASTM D638
Flexural Modulus	1100 MPa	ASTM D790
Notched Izod Impact	110 J/m	ASTM D256

This profile reflects a balanced stiffness–impact combination typical of impact modified PP.

5. Hardness

- Rockwell Hardness (R Scale): 105 (ASTM D785)

6. Thermal Properties

Property Value Notes

DTUL (66 psi, unannealed) Listed in full datasheet Requires full Prospector access

General Thermal Behavior Suitable for standard PP molding temperatures —

7. Processing Guidelines

- Primary Method: Injection molding
- Applications: Caps, closures, toys, thin walled consumer goods, sporting goods
- Form: Pellets

Item No. Supplies/Service Quantity U/M Unit Price Total Price ARO Date

0002	FSPOLYPROP00	5,000	LBS			
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POLY PRO FDA NATURAL PELLETS

Item No. Supplies/Service Quantity U/M Unit Price Total Price ARO Date

0003	FSPOLYPROP01	1,000	LBS			
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POLY PRO FDA TAN PELLETS

Item No. Supplies/Service Quantity U/M Unit Price Total Price ARO Date

0004	FSPOLYPROP02	500	LBS			
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POLY PRO FDA ORANGE PELLETS

Item No. Supplies/Service Quantity U/M Unit Price Total Price ARO Date

0005	FSPOLYPROP04	500	LBS			
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POLY PRO FDA AMBER PELLETS

Material Overview

Polypro P00 is a polypropylene random copolymer (PPR) designed for pressurized liquid-handling systems. It is supplied as natural, unpigmented resin pellets suitable for extrusion and injection molding. The material offers high impact strength, excellent weldability, and long-term thermal stability, making it suitable for potable-water and industrial piping applications.

Physical Properties

- **Material Type** — Polypropylene Random Copolymer (PPR)

- **Color** — Natural / unpigmented
- **Density** — $\sim 0.90 \text{ g/cm}^3$ (ISO 1183)
- **Surface Tension** — $30\text{--}35 \text{ mJ/m}^2$
- **Linear Expansion Coefficient** — $0.16\text{--}0.18 \text{ mm/m}\cdot\text{K}$ (DIN 53752)
- **Thermal Conductivity** — $0.23 \text{ W/m}\cdot\text{K}$ (EN 12664)
- **Surface Resistivity** — $>10^{16} \Omega$ (IEC 60093)

Mechanical Properties

- **Elastic Modulus (E-Modulus)** — $\sim 900 \text{ N/mm}^2$ (ISO 527)
- **Impact Strength** — High, suitable for pressurized systems
- **Weldability** — Excellent for fusion-welded piping
- **Hardness** — Typical of PPR grades, balancing durability and flexibility

Thermal Performance

- **Operating Temperature Range** — 0°C to 80°C (32°F to 176°F)
- **Recommended Valve Temperature Limit** — $<60^\circ\text{C}$
- **Thermal Cycling Resistance** — Suitable for hot and cold water systems

Chemical Resistance

- Resistant to most acids, bases, salts, and organic compounds
- Non-corrosive and non-toxic
- Suitable for potable water applications
- Maintains stability under long-term water exposure

Manufacturing & Form

- **Form** — Cylindrical resin pellets
- **Packaging** — Typically 25 kg bags
- **Processing Methods** —
 - Extrusion (pipes)
 - Injection molding (fittings and valves)

- **Inner Surface Roughness** — $Ra \leq 1 \mu m$ for low-friction flow and reduced scaling

Common Applications

- Injection-molded fittings and valves

Compliance & Standards

- Meets typical PPR requirements for potable-water systems
- Compatible with ISO and ASTM standards for polypropylene materials
- Suitable for use in systems requiring long-term hydrostatic strength